

ION - TESTER

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With the ever-increasing demand for better food, safer automobiles and cleaner air, new products are being produced almost as rapidly as the proverbial hare with his prolific capabilities. The state of the art is proceeding like the tides. Each day brings something new and exciting.

At GOLDEN ENTERPRISES we are pleased to announce our latest breakthrough in creative design which we call the ION-TESTER. This clever device consists of a unique circuit design that has the ability to sense minute ion charges and display these tiny charges on a panel meter. For years, scientists have known that these tiny charges have a definite influence on our bodily functions and well being. The positively charged particles are not conducive to well being, while on the other hand, the negatively charged particles have a way of making us feel good. While all the information is not yet in, scientists are hard at work discovering more each day of the whys and wherefor of these mysterious ions.

The ION-TESTER is able to detect the presence of negative ions.* This small, but extremely sensitive meter, can detect a charge on a pocket comb at a distance of six feet. Its input impedance is exceptionally high, allowing the instrument to detect minute currents.

The ION-TESTER is a complement device to our negative ion generators. (Golden Enterprises Plan #103 (ION-AIR) and our Model #115 (MINI-AIR). With the ION-TESTER you can check to verify that your negative ion generator is indeed producing negative ions. This gives you a ready indicator that can provide a 24-hour surveillance on your ion generator, keeping you informed as to the productivity of negative ions.

There are several other uses for this unique tester. The following is only a partial list of possible applications:

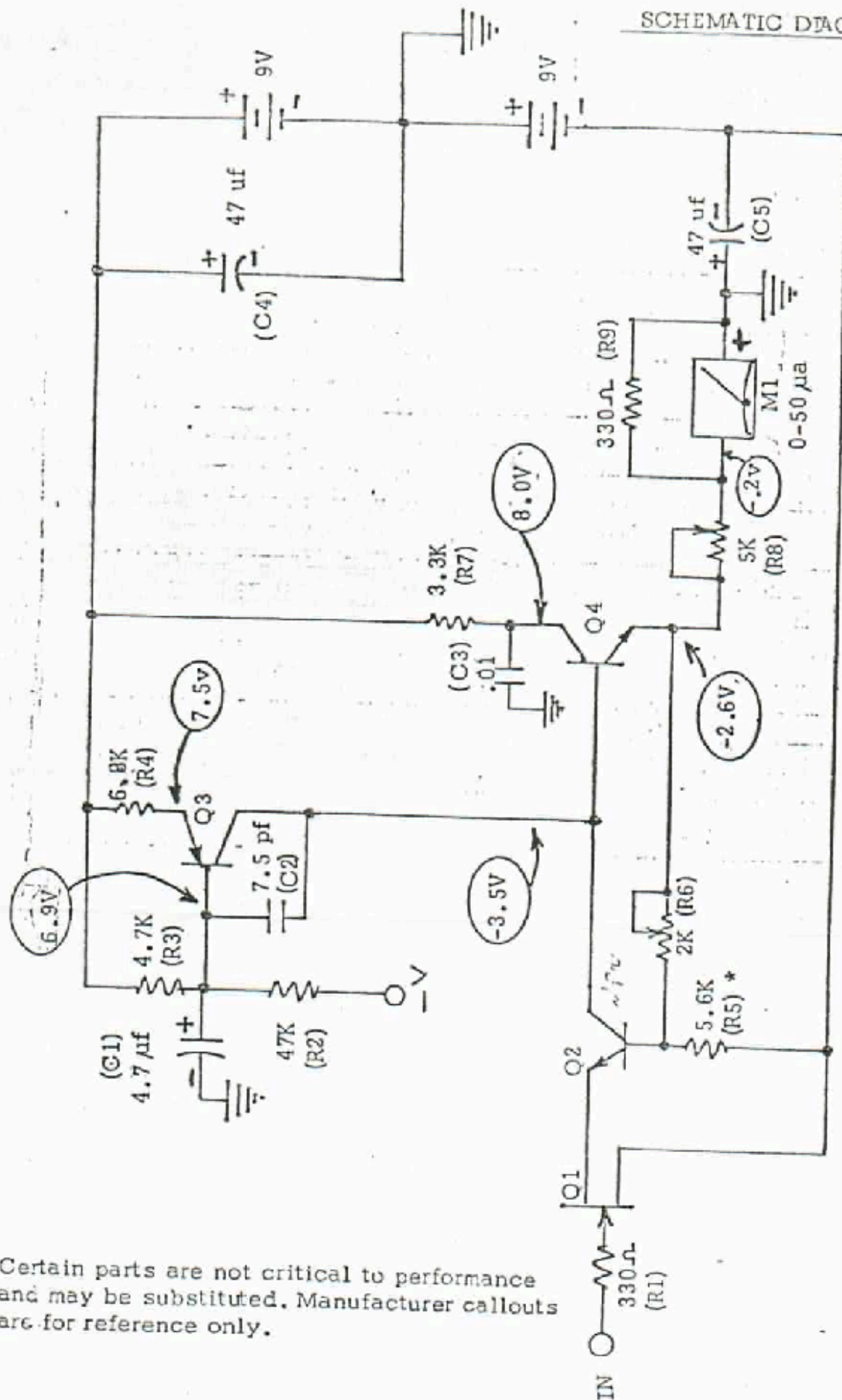
1. Checking your laundry for static electricity
2. Static buildup on carpets, drapes, etc.
3. Observing storm fronts as they move in

This list is only limited by ones own imagination.

The ION-TESTER consists of four (4) readily available solid state devices. The input stage (Q1) is a FET. This is the heart of our mighty little tester! Its input impedance is nearly 1,000 million ohms (1,000,000,000 ohms), or 10^9 ohms. At these input impedance levels, current becomes more important than voltage. We are now sensing tiny currents on charged particles. Q2 is a common base stage driving a common emitter stage (Q4). A constant current source is used (Q3) to stabilize the drift currents. On a 24-hour test, the drift was 2% of the total meter scale. This is negligible and can be easily corrected by resetting the meter zero with R6. The transistors used are the garden variety type and inexpensive... and basically, THAT'S IT!

* GOLDEN ENTERPRISES offers another exciting device called the SUPER-TESTER, Model No. 112A, which tests for the presence of both negative and positive ions.

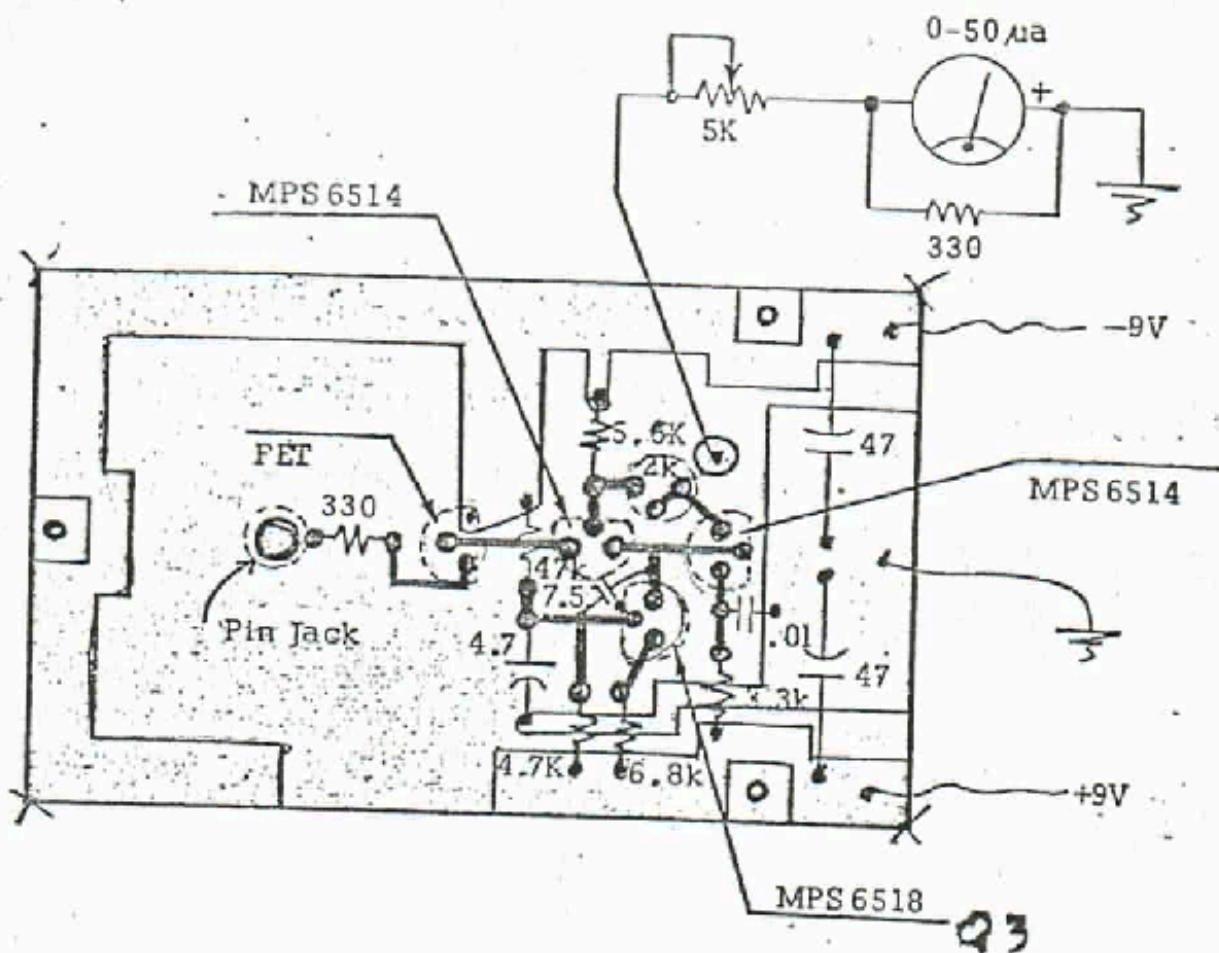
SCHEMATIC DIAGRAM



* MAY BE NECESSARY TO ADJUST

NOTE: Certain parts are not critical to performance and may be substituted. Manufacturer callouts are for reference only.

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PRINTED CIRCUIT BOARD LAYOUT

PARTS LIST

Below is the complete parts list for the ION-TESTER, current as of February 1, 1977. All parts are available from GOLDEN ENTERPRISES for the prices noted, plus shipping and handling charges.

GOLDEN ENTERPRISES offers the ION-TESTER in complete kit form (all parts listed below) for the discount price of \$40.00...\$5.00 additional shipping.

NOTE: All resistors are 1/4 watt unless otherwise specified.

<u>Item No.</u>		<u>Description</u>	<u>Price</u>
1		Printed Circuit Board	5.00
2	R1	330 ohm resistor	.30
3	R2	47K ohm resistor	.30
4	R3	4.7K ohm resistor	.30
5	R4	6.8K ohm resistor	.30
6	R5	5.6K ohm resistor	.30
7	R6	2K (Pot)	3.00
8	R7	3.3K ohm resistor	.30
9	R8	5K (Pot)	3.00
10	R9	330 ohm resistor	.30
11	C1	4.7 uf/25V capacitor	1.00
12	C2	7.5 pf/disc capacitor	.30
13	C3	.01 uf capacitor	.30
14	C4	47 uf capacitor	.50
15	C5	47 uf capacitor	.50
16	Q1	2N5716 (FET) Motorola	2.25
17	Q2	MPS6514 transistor (Motorola) = Sec 15 NPN	2.00
18	Q3	MPS6518 transistor (Motorola) = Sec 19 PNP	2.00
19	Q4	MPS6514 transistor (Motorola)	2.00
20	M1	0-50 ua Meter (Radio Shack 22-051 or equivalent)	12.00
21		Glamour Cabinet (or any suitable housing approximately 4"x 6"x 3")	8.50
			\$44.45

GOLDEN ENTERPRISES offers a completely assembled and tested ION-TESTER for \$75.00 (shipping and handling \$5.00 additional.)

NOTE: Certain parts are not critical to performance and may be substituted. Manufacturer callouts are for reference only.

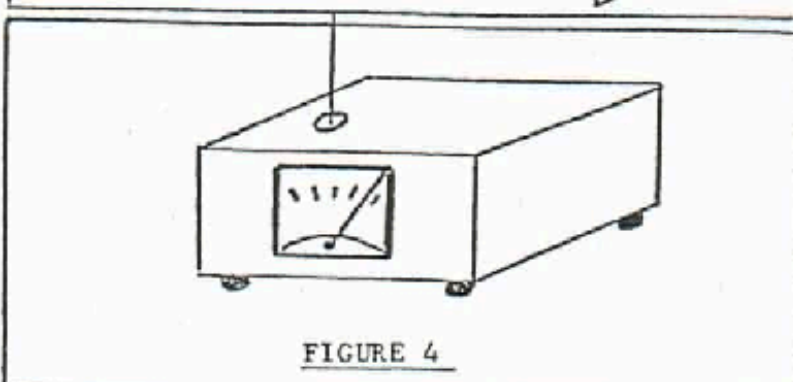
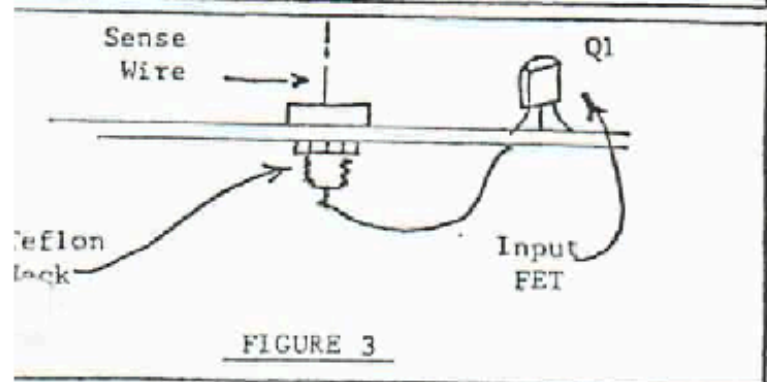
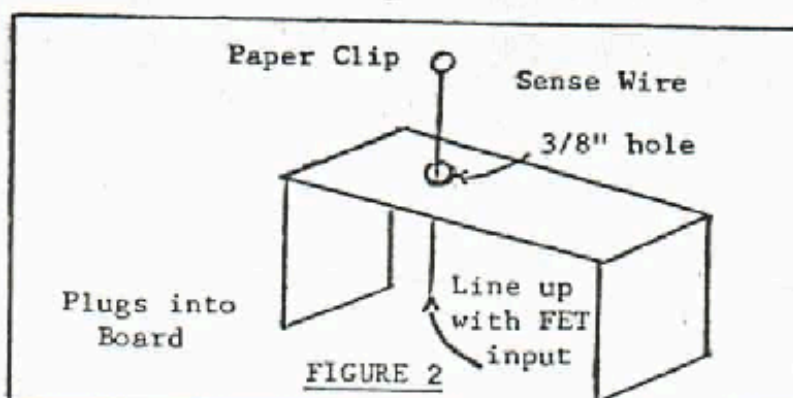
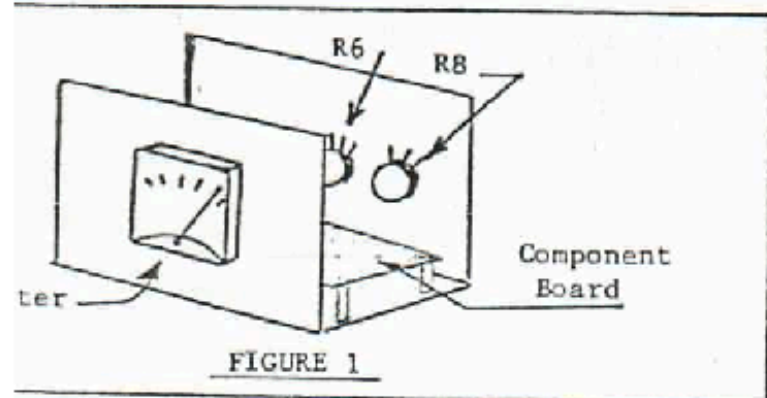
CONSTRUCTION PLANS

The ION-TESTER can be constructed for about \$40.00, using readily available components. To assist you in building the project, you can order a printed circuit board, ready to assemble, from Golden Enterprises for \$5.00. The ION-TESTER is also available in complete kit form for \$40.00.

The ION-TESTER may be constructed in any small, convenient housing. Our unit was put into a 4" x 6" x 3" glamour cabinet. This worked out very well and the following instructions are based upon using this type of housing.

The housing comes in two pieces. The meter mounting hole is cut in the front panel with a one-inch chassis punch. For the next step, a pair of aircraft tin snips is used to cut the hole to proper size to accommodate the panel meter. See Figure 1. Drill the holes to mount R6 and R8 and put them on the back panel, as they are only used occasionally.

The circuit board is mounted on 3/4" to 1" standoffs or spacers to elevate it as high as possible. See Figure 1. This board is mounted on the same part of the housing as the meter. A piece of double-sided masking tape serves as the battery holders. Once the wiring is completed, put the top part of the chassis over the main frame (only temporarily). The input to the FET (Q1) is noted. Directly above this, drill a 3/8" hole in the top part of the chassis. It is very important for this to line up with the input to the FET. The input to Q1 should be a teflon jack as shown in Figure 3 below. The sense wire will plug into this jack. A paper clip unfolded so that it is straight is plugged into this jack. This should be positioned so that it falls in the center of the 3/8" hole in the upper part of the housing as shown in Figure 2.



Once the ION-TESTER is assembled and wired, it can be tested without the upper cover of the housing.

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INSTRUCTIONS FOR USE

When using your ION-TESTER, it is important to remember the following points:

1. When taking reference readings, set the tester approximately 16" from the measuring source on a steady table.
2. For spot checking, hold the ION-TESTER in the palm of your hand extended at arms length so that your own charge does not influence the reading.
3. The meter reads DOWNSCALE and is normally (in the absence of negative ions) at 50 or slightly above.
4. There is no ON-OFF switch because the battery life will be approximately equal to the shelf life. This is due to the fact that a very tiny current is consumed by the tester.
5. The tester does not indicate absolute quantities, but indicates the presence of negative ions. When the ION-TESTER is exposed to a flow of negative ions (such as from the ION-AIR negative ion generator) the meter needle will read downscale.
6. When measuring the ions from your ION-AIR negative ion generator, you may expect a slight fluctuation in the meter reading. This is normal, as the flow of ions is not at a perfectly steady rate.
7. The small wire extending upward is the sensing wire and care should be taken so that it does not touch the housing.

GOLDEN ENTERPRISES neither expresses nor implies any guaranty other than the circuits and construction plans described are correct and the unit will operate properly when constructed according to instructions. We do not assume any responsibility for any incidents arising from misuse, or negligence. Builders and/or user assumes full responsibility.

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TESTING INSTRUCTIONS

Connect the two nine-volt batteries and with a good volt ohmmeter, measure the voltage on the Q2, Q3 collector at the base of Q4. All these points are common to each other. It should measure near -3.65 volts. Check the schematic for the other voltages. If the voltages check out, you can now check the activity and sensitivity. Collect the following items from around the house: Pocket comb, plastic produce bag, and a silk cloth.

Adjust R8 for the maximum down scale reading (about 80 on the meter). Next, adjust R6 to its mid-position (it should move about 4 divisions). Now, re-adjust R8 to 100 on the meter scale. Your meter is now ready to use.

To test, perform the following procedures:

Run your pocket comb through your hair, then point it at the meter while standing about 3 feet away. The meter should read all the way down scale temporarily. It will soon come back up as the charge bleeds off the comb and into the meter. Move forward slightly and observe that the meter moves down scale again. Now, try the plastic bag. Rub it vigorously over your clothing and then point it at your ION-TESTER. Once again, the meter moves down scale, indicating the presence of negative ions. Do the same with the silk cloth.

As you become more familiar with your ION-TESTER, you will think of many other interesting applications.

Should you experience any difficulty with the unit, check the following:

1. Center the sense wire in the 3/8" hole. Keep away from metal parts.
2. Clean the teflon jack with alcohol.
3. Make sure you are using fresh batteries.
4. Check for correct meter polarity.
5. Make sure all the voltage readings are correct $\pm$ 10%.
6. All solder connections must be good.

If you have carefully followed the instructions and schematic, your ION-TESTER should work as described. It will give you many hours of fascinating entertainment, as well as being a truly scientific instrument. If you wish a more sophisticated device, one that will indicate both positive and negative ions, send for our SUPER TESTER, Plan 112A.

If on studying these instructions, you decide not to build the Ion-Tester yourself, you may purchase one completely assembled, tested and ready to use from GOLDEN ENTERPRISES for \$75.00. It is also available in kit form for \$40.00. . . (\$5.00 additional for shipping and handling)